US 20250284771A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0284771 A1
(43) Pub. Date: Sep. 11, 2025
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(54) GENERATION OF A UNIFORMLY RANDOM VECTOR

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(21) Appl. No.: **19/060,119**

(22) Filed: **Feb. 21, 2025**

(30) **Foreign Application Priority Data**

Mar. 8, 2024 (FR) FR2402329

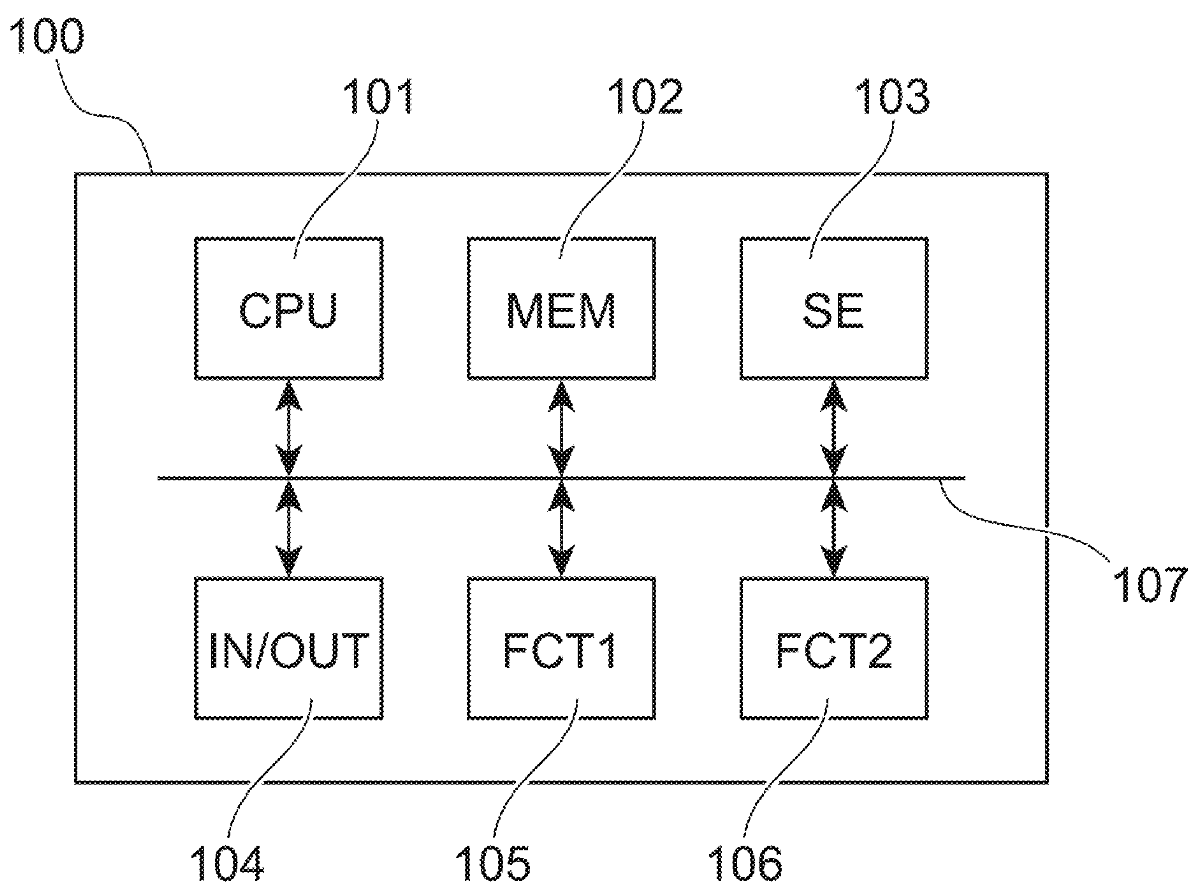
Publication Classification

(51) **Int. Cl.**
G06F 17/16 (2006.01)
H04L 9/32 (2006.01)

(52) **U.S. Cl.**
CPC *G06F 17/16* (2013.01); *H04L 9/3247* (2013.01)

(57) **ABSTRACT**

The present disclosure relates to a generation method, adapted to be executed by an electronic device, of an first uniformly random vector comprising n first elements, n being an integer equal or greater than two, comprising generating a second uniformly random vector comprising n second elements; generating m third elements, m being an integer equal or greater than one; generating a third uniformly random vectors comprising 1 third elements and n−1 second elements, and having third elements $(b_0, \ldots, b_{n-1})$ different from the second elements at the same position in the second uniformly random vector and the other third uniformly random vectors; and calculating the first elements by applying an unmasking function to the second elements and to the m third elements.



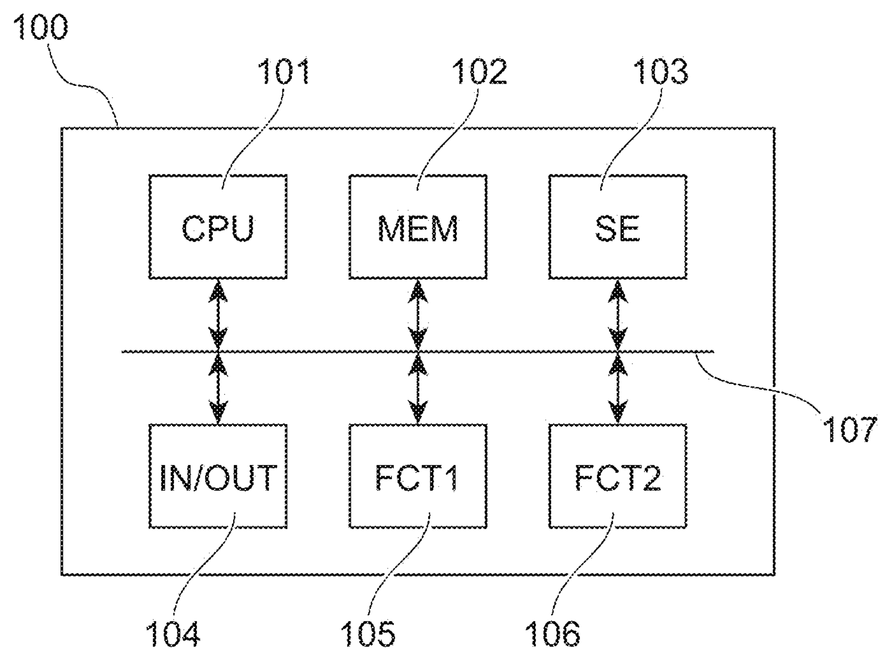

100
101
102
103
CPU
MEM
SE
107
IN/OUT
FCT1
FCT2
104
105
106

FIG. 1

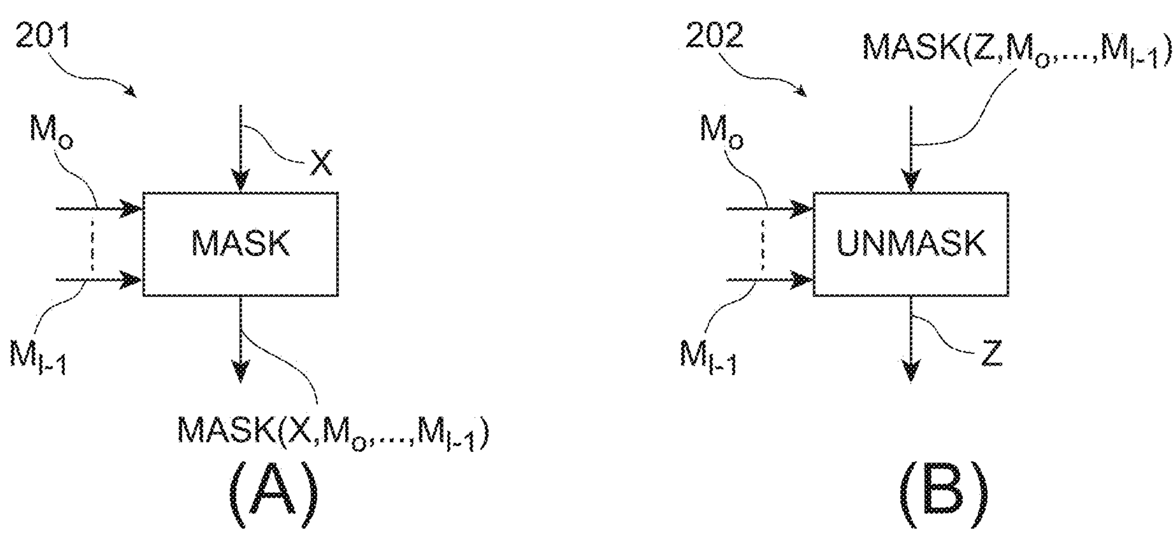
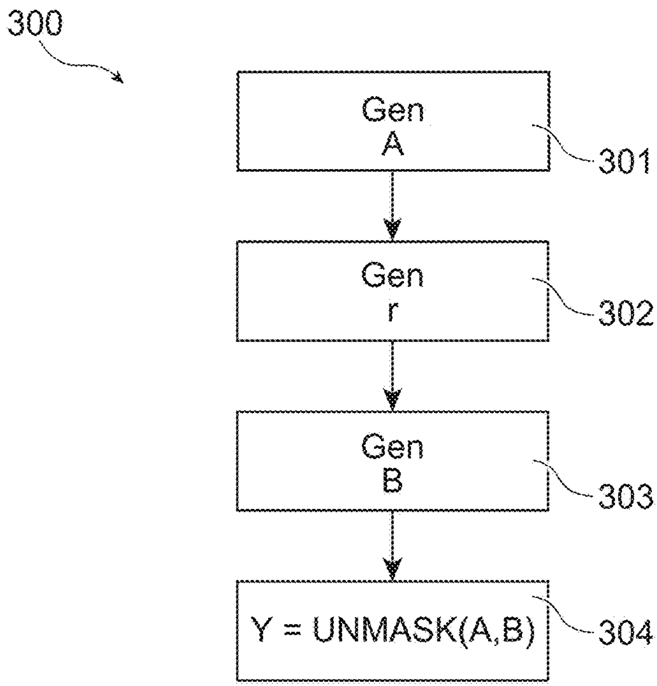
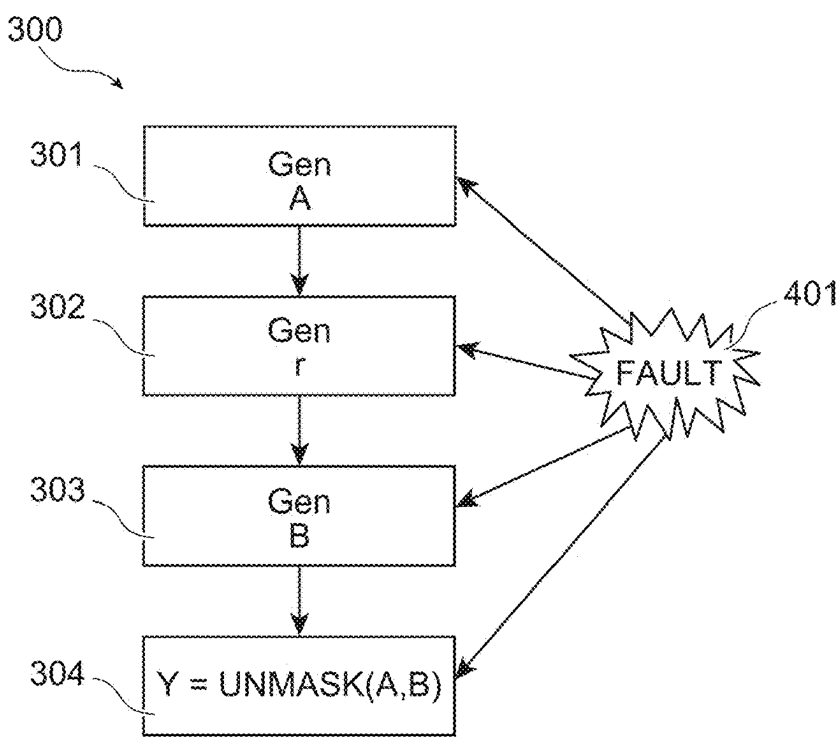

201
M_o
X
MASK
M_{l-1}
$MASK(X,M_o,\ldots,M_{l-1})$
(A)
202
$MASK(Z,M_o,\ldots,M_{l-1})$
M_o
UNMASK
M_{l-1}
Z
(B)

FIG. 2



300
Gen
A
301
Gen
r
302
Gen
B
303
Y = UNMASK(A,B)
304

FIG. 3



300
301
Gen
A
302
Gen
r
303
Gen
B
304
Y = UNMASK(A,B)
FAULT
401

FIG. 4

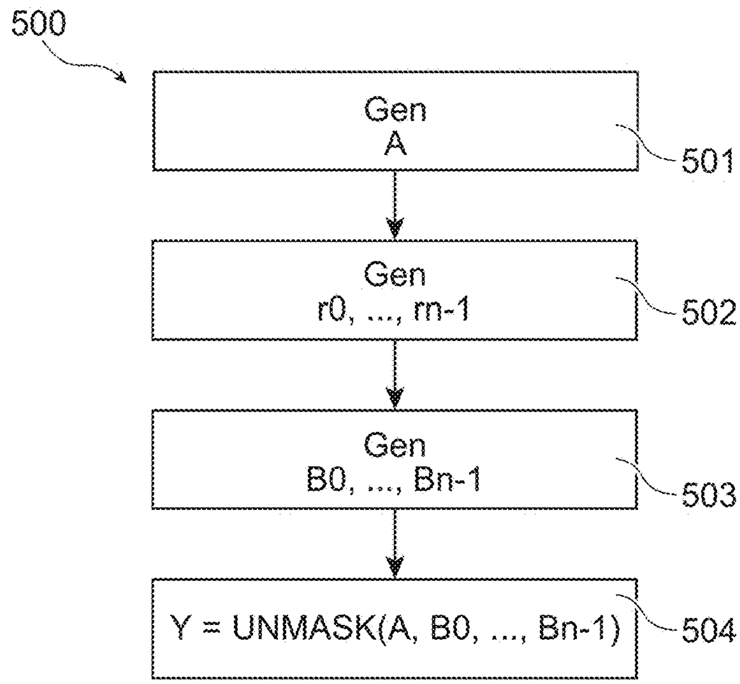

500
Gen
A
501
Gen
r0, ..., rn-1
502
Gen
B0, ..., Bn-1
503
Y = UNMASK(A, B0, ..., Bn-1)
504

FIG. 5

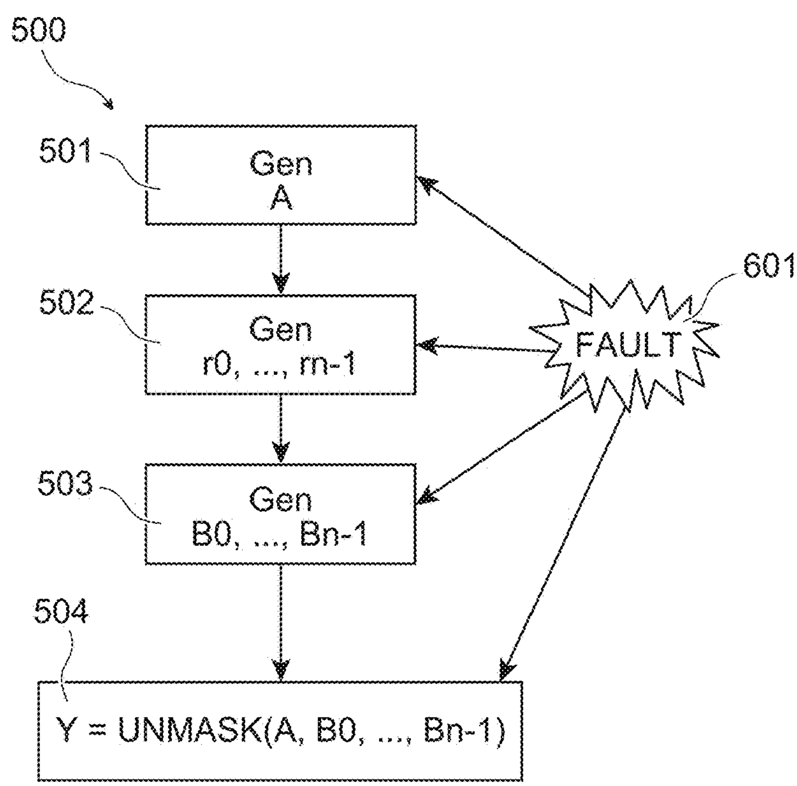

500
501
Gen
A
502
Gen
r0, ..., rn-1
503
Gen
B0, ..., Bn-1
601
FAULT
504
Y = UNMASK(A, B0, ..., Bn-1)

FIG. 6

GENERATION OF A UNIFORMLY RANDOM VECTOR

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the priority benefit of French Patent Application Number FR2402329, filed on Mar. 8, 2024, entitled "Génération d'un vecteur uniformément aléatoire", which is hereby incorporated by reference to the maximum extent allowable by law.

TECHNICAL FIELD

[0002] The present disclosure relates generally to electronic systems and devices, and more particularly to the security of such systems and devices. The present disclosure relates, more precisely, to cryptographic methods and to their implementation.

BACKGROUND

[0003] Different techniques for authenticating and/or ensuring the integrity of secret and/or sensitive data are used nowadays. Digital signatures are one of them, and consist of the application of one or more cryptographic algorithms to data, such as sensitive data. Many cryptographic algorithms, and more especially many asymmetric cryptographic algorithms use uniformly random data.

[0004] In certain cases, these data are generated periodically, for example each time a data signature process is implemented. It may then be necessary to secure the generation of these uniformly random data.

[0005] It would be desirable to be able to improve at least in part certain aspects of the generation of uniformly random data.

BRIEF SUMMARY

[0006] There is a need for more robust generation method of uniformly random data, and, in particular, uniformly random vector.

[0007] There is a need for generation method of uniformly random data, and, in particular, uniformly random vector, that are secured against single fault injection attacks and multiple faults' injection attacks.

[0008] There is a need for an electronic device capable of executing such a generation method.

[0009] One embodiment addresses all or some of the drawbacks of known generation methods of uniformly random data, and, in particular, known generation methods of uniformly random vector.

[0010] One embodiment provides a generation method of uniformly random data, and, in particular, a uniformly random vector.

[0011] One embodiment provides an electronic device capable of implementing such a generation method.

[0012] One embodiment provides a generation method, adapted to be executed by an electronic device, of an first uniformly random vector comprising n first elements, n being an integer equal or greater than two, each first element being comprised in a finite set T, comprising the following steps: generate a second uniformly random vector comprising n second elements; generate m third elements, m being an integer equal or greater than one; generate m third uniformly random vectors ranked from zero to m−1, wherein third uniformly random vector ranked k, k being an integer comprised between zero and m−1, comprises k third elements and n-k second elements, each third uniformly random vector having third elements different from the second elements at the same position in the second uniformly random vector and the other third uniformly random vectors; and calculate said first elements by applying an unmasking function to said second elements and to said m third elements.

[0013] Another embodiment provides an electronic device capable of implementing a generation method of a uniformly random first vector comprising n first elements, n being an integer equal or greater than two, each first element being comprised in a finite set T, comprising the following steps: generate a second uniformly random vector comprising n second elements; generate m third elements, m being an integer equal or greater than one; generate m third uniformly random vectors ranked from zero to m−1, wherein third uniformly random vector ranked k, k being an integer comprised between zero and m−1, comprises k third elements and n-k second elements, each third uniformly random vector having third elements different from the second elements at the same position in the second uniformly random vector and the other third uniformly random vectors; and calculate said first elements by applying an unmasking function to said second elements and to said m third elements.

[0014] According to an embodiment, the generation step of said third uniformly random vector ranked k comprises the following steps: use the n first elements; shift said n first elements of at least k steps; and replace at least k elements from the first elements by k elements from the third elements.

[0015] According to an embodiment, each first, second and third elements are chosen in a finite set.

[0016] According to an embodiment, said first vector is a commitment vector of the Dilithium algorithm.

[0017] Another embodiment provides an execution method of a cryptographic algorithm comprising said generation method described previously.

[0018] According to an embodiment, said cryptographic algorithm is a signature method.

[0019] According to an embodiment, said cryptographic algorithm is the Dilithium algorithm.

BRIEF DESCRIPTION OF DRAWINGS

[0020] The foregoing features and advantages, as well as others, will be described in detail in the following description of specific embodiments given by way of illustration and not limitation with reference to the accompanying drawings, in which:

[0021] FIG. **1** illustrates, schematically and in block form, an embodiment of an electronic device configured to execute the methods of FIGS. **3** and **5**;

[0022] FIGS. **2**A and **2**B illustrate, schematically and in block form, a masking function and an unmasking function;

[0023] FIG. **3** illustrates, in block form, an embodiment of a generation method of a uniformly random vector;

[0024] FIG. **4** illustrates, in block form, a fault injection attack upon the embodiment of FIG. **3**;

[0025] FIG. **5** illustrates, in block form, another embodiment of a generation method of a uniformly random vector; and

[0026] FIG. **6** illustrates, in block form, a fault injection attack upon the embodiment of FIG. **5**.

DETAILED DESCRIPTION

[0027] Like features have been designated by like references in the various figures. In particular, the structural and/or functional features that are common among the various embodiments may have the same references and may dispose identical structural, dimensional and material properties.

[0028] For the sake of clarity, only the operations and elements that are useful for an understanding of the embodiments described herein have been illustrated and described in detail. In particular, cryptographic algorithms using the embodiments described hereafter are not detailed hereafter. These implements of cryptographic algorithms are obvious for the person skilled in the art based on the following disclosure.

[0029] Unless indicated otherwise, when reference is made to two elements connected together, this signifies a direct connection without any intermediate elements other than conductors, and when reference is made to two elements coupled together, this signifies that these two elements can be connected or they can be coupled via one or more other elements.

[0030] In the following disclosure, unless indicated otherwise, when reference is made to absolute positional qualifiers, such as the terms "front", "back", "top", "bottom", "left", "right", etc., or to relative positional qualifiers, such as the terms "above", "below", "higher", "lower", etc., or to qualifiers of orientation, such as "horizontal", "vertical", etc., reference is made to the orientation shown in the figures.

[0031] Unless specified otherwise, the expressions "around", "approximately", "substantially" and "in the order of" signify within 10%, and preferably within 5%.

[0032] The above described embodiments concern the generation of a uniformly random data, and more particularly the generation of a uniformly random vector. Such a data or vector can be used during the execution of a cryptographic algorithm, such as a signature algorithm. Moreover, the above described embodiments concern a generation method, and the electronic device executing it, that is resistant to single and multiple fault injection attacks.

[0033] Moreover, the above described embodiments are particularly adapted to the generation of uniformly random vectors that are used by cryptographic algorithm such as any signature algorithm that is not deterministic, for example the Dilithium algorithm, the ECDSA algorithm or the qTESLA algorithm.

[0034] FIG. **1** shows, very schematically and in block form, an electronic device **100** capable of implementing the generation method of a uniformly random vector described in relation with FIGS. **3** to **6**.

[0035] Device **100** is an electronic device adapted to processing data, and more particularly, to execute algorithm using uniformly random data, and to generate such uniformly random data. For example, device **100** is capable of executing cryptographic algorithm.

[0036] Device **100** comprises a processor **101** (CPU) capable of processing data. According to an example, device **100** may comprise a plurality of processors, each adapted to processing different types of data. According to a specific example, device **100** may comprise at least one processor that is adapted to execute algorithm using uniformly random data, and to generate such uniformly random data.

[0037] Device **100** further comprises one or a plurality of memories **102** (MEM) into which are stored data, for example, critical data. According to an example, device **100** comprises a plurality of types of memories, such as a ROM, a RAM, a volatile memory, and/or a non-volatile memory. According to a specific example, device **100** may comprise one or more memories that are trusted by the processor **101** and that have a secure access to the processor **101**. According to an embodiment, device **100** comprises one or more external memories, or untrusted memories, that do not have a secure access to the processor **101**.

[0038] Device **100** further comprises one or more secure element **103** (SE) adapted to processing critical and/or secret data. Secure element **103** may comprise its own processor (s), its own memory or memories, etc. According to an example, the one or more secure elements **103** are capable of executing algorithm using uniformly random data, and generating uniformly random data.

[0039] Device **100** further and optionally comprises one or a plurality of input/output circuits **104** (I/O) enabling device **100** to transmit and/or to receive data and/or energy with one or a plurality of external electronic devices.

[0040] Device **100** further comprises one or a plurality of circuits **105** (FCT**1**) and **106** (FCT**2**) implementing one or a plurality of functionalities of device **100**. According to an example, circuits **105** and **106** may comprise specific data processing circuits, such as signature generation circuits, or circuits enabling to perform measurements, such as sensors. According to an example, the circuits **105** and **106** are capable of executing algorithm using uniformly random data, and generating uniformly random data.

[0041] Device **100** further comprises one or a plurality of communication buses **107** enabling all the circuits of device **100** to communicate. In FIG. **1**, a single bus **107** coupling processor **101**, memory or memories **102**, secure element **103**, and circuits **104** to **106** is shown, but in practice, device **100** comprises a plurality of communication buses coupling these different elements.

[0042] FIG. **2** comprises two views (A) and (B) each representing, very schematically and in block forms, respectively, a masking function **201** and an unmasking function **202**.

[0043] Masking function **201** (MASK) takes at least two types of inputs: a data X and l masks $M_0, \ldots, M_{l-1}$, l being an integer greater than or equal to one (1), and provides a masked data MASK $(X, M_0, \ldots, M_{l-1})$. Data X, masks $M_0, \ldots, M_{l-1}$, and masked data MASK $(X, M_0, \ldots, M_{l-1})$ are of the same nature, such as vector comprising the same number of elements. Masking function **201** is constructed in order to provide a masked data MASK$(X, M_0, \ldots, M_{l-1})$ that is statistically independent from the data X and that is statistically independent from masks $M_0, \ldots, M_{l-1}$.

[0044] Unmasking function **202** (UNMASK) takes at least two types of inputs: a masked data Z and l masks $M_0, \ldots, M_{l-1}$, and provides an unmasked data UNMASK $(Z, M_0, \ldots, M_{l-1})$. Masked data Z, masks $M_0, \ldots, M_{l-1}$ and unmasked data UNMASK$(Z, M_0, \ldots, M_{l-1})$ are of the same nature, such as vector comprising the same number of elements. Unmasking function **202** is constructed such that for fixed masks M**0**, . . . , Ml−1 the composition of the masking and the unmasking functions acts like the identity function. In other words, unmasking the masked value MASK $(X, M_0, \ldots, M_{l-1})$ with the masks $M_0, \ldots, M_{l-1}$ gives back data X.

[0045] Furthermore, the unmasking function **202** can be used for generating a uniformly random data, and especially a uniformly random vector. In this disclosure, it is called a uniformly random data, a data obtained through a method choosing an element of the set such that every element is equally likely to be obtained. Moreover, a uniformly random vector is a vector obtained through a method that independently selects its elements as a uniformly random data.

[0046] If the masked data Z and/or masks $M_0, \ldots, M_{l-1}$ have been proven to be uniformly random and are independent from each other, then the unmasked data UNMASK(Z, $M_0, \ldots, M_{l-1}$) is also uniformly random. Thus, choosing l+1 uniformly random data and applying to them the unmasking function **202** is a way to generate another uniformly random data. This property is used in the embodiments described in relation with FIGS. **3** to **6**.

[0047] The masking mechanism is well-known, and practical examples of this mechanism are the Boolean masking and the arithmetic masking. Other kind of masking mechanisms are obvious for the person skilled in the art.

[0048] FIG. **3** is a block diagram representing a first embodiment of a generation method **300** of a uniformly random vector Y.

[0049] The uniformly random vector Y comprises n elements $y_0, \ldots, y_{n-1}$, each element $y_0, \ldots, y_{n-1}$ being from a finite set T. n is an integer greater or equal to two. Thus, the vector Y can be seen as an element of the finite set T^n.

[0050] According to an example, vector Y can be a commitment vector of the Dilithium algorithm.

[0051] At an initial step **301** (Gen A), a uniformly random vector A from finite set T^n is generated. Vector A comprises n elements $a_0, \ldots, a_{n-1}$. In other words, at step **301** n uniformly random elements $a_0, \ldots, a_{n-1}$ from set T are generated and are used to form the uniformly random vector A. According to an embodiment, elements $a_0, \ldots, a_{n-1}$ are all independent from each other.

[0052] At a step **302** (Gen r), following step **301**, another uniformly random element r from set T is generated. According to an embodiment, element r is independent from each element $a_0, \ldots, a_{n-1}$.

[0053] At a step **303** (Gen B), following step **302**, a uniformly random vector B from finite set T^n is constructed by using the element r and only n−1 elements from elements $a_0, \ldots, a_{n-1}$. Vector B comprises n elements $b_0, \ldots, b_{n-1}$.

[0054] Vector B is obtained by shuffling elements $a_0, \ldots, a_{n-1}$ and replacing one of these elements by element r. According to an embodiment, all elements that are common from vectors A and B have a different place.

[0055] According to a preferred embodiment, vector B is obtained by shifting the position of elements $a_0, \ldots, a_{n-1}$ of at least one step, and then by replacing one of these elements by element r. According to a practical example, vector B can be given by the following mathematical formula:

$$B = (b_0, \ldots, b_{n-1}) = (r, a_0, \ldots, a_{n-2}) \qquad \text{[Math 1]}$$

[0056] In this case, vector B is obtained by shifting the position of elements $a_0, \ldots, a_{n-1}$ of one step, and by replacing element a_{n-1} by element r.

[0057] Each element b_i of vector B is independent from the element a_i of vector A.

[0058] At a step **304** (Y=UNMASK(A,B)), following step **303**, the uniformly random vector Y is generated by using vectors A and B, and the unmasking function **202** described in relation with FIG. **2** wherein 1 is equal to one (1). Each element y_i of the vector Y is given by the following mathematical formula:

$$y_i = \text{UNMASK}(a_i, b_i) \qquad \text{[Math 2]}$$

[0059] Elements a_i and b_i are independent and uniformly random, thus, element y_i is also uniformly random. Since the elements $y_0, \ldots, y_{n-1}$ are independent and uniformly random, Y is a uniformly random vector.

[0060] A first advantage of this generation method is that this method is resistant to single fault injection attacks. This advantage is described in more details in relation with FIG. **4**.

[0061] A second advantage of this generation method is that it uses the generation of only n+1 uniformly random elements even though, by using the unmasking function **202**. A naïve solution based on the unmasking function **202** that is robust against a single fault injection would require 2*n uniformly random elements.

[0062] FIG. **4** represents a block diagram illustrating a single fault injection attack upon the generation method of FIG. **3**.

[0063] Fault injection attacks are a class of physical attacks that can be conducted against a device, such as device **100** of FIG. **1**, in order to modify the way the device operates. These types of attacks are mostly used for malicious purposes such as extraction of sensitive data. During a single fault injection attack, a person can modify the value of a data, and more particularly, can modify the value of several bits forming said data. During a multiple faults' injection attack, a person can modify the value of several data, and more particularly, can force the value of several bits forming said several data.

[0064] Concerning the generation method **300** of FIG. **3**, a single fault injection attack can occur during any step of the method. On other words, a fault **401** (FAULT) can be injected at any step of the method **300**.

[0065] Concerning the generation method **300**, a single fault injection attack can modify the value of vector A. It means that one element a_i of vector A has been replaced by another element t that is not uniformly random. Thus, vector A is given by the following mathematical formula:

$$A = (a_0, \ldots, a_i, \ldots, a_{n-1}) = (a_0, \ldots, t, \ldots, a_{n-1}) \qquad \text{[Math 3]}$$

[0066] Such a fault can be injected at any step of method **300**. When this fault is injected before **303**, vector B can be potentially generated by using the element t. However, as said earlier, all elements that are common from vector A and B have a different place. Thus, each element $y_0, \ldots, y_{n-1}$ of vector Y is given by the result of the application of the unmasking function **202** to a uniformly random element and another independent element that is not necessarily uniformly random. Vector Y is thus still comprising only uniformly random and independent elements.

[0067] According to the practical example of the preferred embodiment described before, vector B is given by the following mathematical formula:

$$B=(b_0,\ldots b_{i+1},\ldots,b_{n-1})=(r,a_0,\ldots,a_i,\ldots,a_{n-2})=(r,a_0,\ldots,t,\ldots,a_{n-2}) \quad \text{[Math 4]}$$

[0068] Element t is placed at the place of element b_{i+1} of vector B. Element y_{i+1} is given by the result of the application of unmasking function **202** to the element b_{i+1}=t and the uniformly random element a_{i+1}. Thus, element y_{i+1} is also uniformly random, and the single fault injection attack does not have an exploitable result.

[0069] When a single fault injection attack modifies the value of vector B, it means that element b_i of vector B, meaning element r or an element of vector A used for generating vector B, has been replaced by the other element t that is not uniformly random.

[0070] According to the practical example of the preferred embodiment described before, vector B is given by the following mathematical formula:

$$B=(b_0,\ldots,b_{n-1})=(t,a_0,\ldots,a_{n-2}) \quad \text{[Math 5]}$$

[0071] In that case, only vector B is impacted by the fault injection. Vector A is still uniformly random. Thus, vector Y is also uniformly random, and the single fault injection attack does not have an exploitable result.

[0072] Furthermore, a fault could be injected to directly modify vector Y, for example, if it is injected during step **304**. However, the person skilled in the art knows that such an attack can be avoided, for example by using any computation on vector Y implemented in a masked fashion by only using the vectors A and B.

[0073] FIG. **5** is a block diagram representing a second embodiment of a generation method **500** of the uniformly random vector Y.

[0074] The generation method **500** is similar to the generation method **300** described in details in relation with FIG. **3**. Common features of methods **300** and **500** are not detailed again hereafter, only their differences are highlighted.

[0075] As said earlier, the uniformly random vector Y comprises n elements $y_0, \ldots, y_{n-1}$, each element $y_0, \ldots, y_{n-1}$ being from the finite set T.

[0076] At an initial step **501** (Gen A), the uniformly random vector A comprising n elements $a_0, \ldots, a_{n-1}$ is generated. According to an embodiment, elements $a_0, \ldots, a_{n-1}$ are all independent from each other.

[0077] At a step **502** (Gen $r_0, \ldots, r_{m-1}$), following step **501**, m uniformly random elements $r_0, \ldots, r_{m-1}$ from set T are generated. m is an integer equal or greater than one (1). According to an embodiment, elements $r_0, \ldots, r_{m-1}$ are each independent from each other and from each element of elements $a_0, \ldots, a_{n-1}$.

[0078] At a step **503** (Gen $B_0, \ldots, B_{m-1}$), following step **502**, m uniformly random vectors $B_0, \ldots, B_{m-1}$ from finite set T^n are constructed by using elements $r_0, \ldots, r_{m-1}$ and elements $a_0, \ldots, a_{n-1}$. Each vector B_k comprises n elements $b_{k,0}, \ldots, b_{k,n-1}$, k being an integer comprised between zero and m−1.

[0079] More particularly, the uniformly random vector B_k ranked k comprises k elements from elements $r_0, \ldots, r_{m-1}$ and n-k elements from elements $a_0, \ldots, a_{n-1}$, and each vector $B_0, \ldots, B_{m-1}$ has its elements disposed in a different order. More particularly, at every position, all the elements from vectors A, $B_0, \ldots, B_{m-1}$ are different.

[0080] Thus, at every position, the elements from vectors A, $B_0, \ldots, B_{m-1}$ are uniformly random and independent from each other.

[0081] According to a preferred embodiment, vector B_k ranked k is obtained by shifting the position of elements $a_0, \ldots, a_{n-k}$ of at least k steps, and then by replacing one of these elements by k elements of elements $r_0, \ldots, r_{m-1}$. According to a practical example, vector B_k can be given by the following mathematical formula:

$$B=(b_0,\ldots b_{k-1},b_k,b_{n-1})=(r_0,\ldots,r_{k-1},a_0,\ldots,a_{n-k-1}) \quad \text{[Math 6]}$$

[0082] In this case, vector Bk is obtained by shifting the position of elements $a_0, \ldots, a_{n-1}$ of k steps, and by replacing element an-k to an−1 by element $r_0, \ldots, r_{k-1}$.

[0083] At a step **504** (Y=UNMASK (A, $B_0, \ldots, B_{m-1}$)), following step **503**, the uniformly random vector Y is generated by using vectors A and $B_0, \ldots, B_{m-1}$, and the unmasking function **202** described in relation with FIG. **2**, wherein 1 is equal to m. Thus, each element y_i, is given by the following mathematical formula:

$$yi=\mathrm{UNMASK}(a_i,b_{0,i},\ldots,b_{m-1,i}) \quad \text{[Math 7]}$$

[0084] Elements $a_i, b_{0,i}, \ldots, b_{m-1,i}$ are independent from each other and uniformly random, thus, element y_i is also uniformly random. Since the elements $y_0, \ldots, y_{n-1}$ are independent and uniformly random, Y is a uniformly random vector.

[0085] FIG. **6** is a block diagram illustrating a multiple faults' injection attack upon the generation method **500** of FIG. **5**.

[0086] As said earlier, fault injection attacks are a class of physical attacks that can be conducted against a device. During a multiple faults' injection attack, a person can modify the value of several data, and more particularly, can modify the value of several bits forming said several data. It is considered here a faults' injection attack wherein m faults are injected.

[0087] Concerning the generation method **300**, a multiple faults' injection attack can modify the value of vector A and/or vector $B_0, \ldots, B_{m-1}$.

[0088] When a multiple faults' injection attack modifies only the value of vector A. It means that m elements a_i of vector A have been replaced by m other elements $t_0, \ldots, t_{m-1}$ that are not uniformly random. Such faults can be injected at any step of method **500**. When this fault is injected before **503**, vectors $B_0, \ldots, B_{m-1}$ can be potentially generated by using some of the m elements $t_0, \ldots, t_{m-1}$. However, as said earlier, all elements that are common from vector A and $B_0, \ldots, B_{m-1}$ have a different place. Thus, each element $y_0, \ldots, y_{n-1}$ of vector Y is given by the result of the application of the unmasking function **202** to a uniformly random element and m another elements that are not necessarily uniformly random. Vector Y is thus still comprising only uniformly random and independent elements.

[0089] According to a preferred embodiment, when m faults are injected during the generation method **500**, at most m elements from $r_0, \ldots, r_k, a_0, \ldots, a_{n-1}$ of vectors B_k are modified. At every position, since the (m+1) elements of the vectors A, $B_0, \ldots, B_{m-1}$ are different and chosen from $r_0, \ldots, r_k, a_0, \ldots, a_{n-1}$, there must be at least one element that is not modified by the faults and is still uniformly random. Thus, each element $y_0, \ldots, y_{n-1}$ of vector Y obtained as the result of the application of the unmasking function **202** to independent elements of which at least one is uniformly random, is still uniformly random. Therefore, vector Y comprising only uniformly random elements is still uniformly random.

[0090] When a multiple faults' injection attack modifies only the value of vectors $B_0, \ldots, B_{m-1}$, it means that elements $b_{0i}, \ldots, b_{m-1i}$ of vectors $B_0, \ldots, B_{m-1}$ have been replaced by other elements $t_0, \ldots, t_{m-1}$ that are not uniformly random. In that case, only vectors $B_0, \ldots, B_{m-1}$ are impacted by the fault injection. Vector A is still uniformly random. Thus, vector Y is also uniformly random, and the multiple faults' injection attack does not have an exploitable result.

[0091] When a multiple faults' injection attack modifies directly the value of vector A and the value of vectors $B_0, \ldots, B_{m-1}$, it means that m elements amongst elements a_i of vector A and elements $b_{0i}, \ldots, b_{m-1i}$ of vectors $B_0, \ldots, B_{m-1}$ have been replaced by m other elements $t_0, \ldots, t_{m-1}$ that are not uniformly random. However, as said earlier, at every position, since the (m+1) elements of the vectors A, B**0**, . . . , Bm−1 are different and chosen from r**0**, . . . , rk, a**0**, . . . , an−1, there must be at least one element that is not modified by the faults and is still uniformly random. Thus, each element y**0**, . . . , yn−1 of vector Y obtained as the result of the application of the unmasking function **202** to independent elements of which at least one is uniformly random, is still uniformly random. Therefore, vector Y comprising only uniformly random elements is still uniformly random.

[0092] Furthermore, faults could be injected to directly modify vector Y, for example, if it is injected during step **504**. However, the person skilled in the art knows that such an attack can be avoided, for example by using any computation on vector Y implemented in a masked fashion by only using the vectors A and $B_0, \ldots, B_{m-1}$.

[0093] Various embodiments and variants have been described. Those skilled in the art will understand that certain features of these embodiments can be combined and other variants will readily occur to those skilled in the art.

[0094] Finally, the practical implementation of the embodiments and variants described herein is within the capabilities of those skilled in the art based on the functional description provided hereinabove.

1. A generation method, adapted to be executed by an electronic device, of a first uniformly random vector comprising n first elements, n being an integer equal or greater than two, each first element being comprised in a finite set T, the method comprising:

generating a second uniformly random vector comprising n second elements;

generating m third elements, m being an integer equal or greater than one;

generating m third uniformly random vectors ranked from zero to m−1, wherein a third uniformly random vector ranked k, k being an integer comprised between zero and m−1, comprises k third elements and n-k second elements, each third uniformly random vector having third elements different from the second elements at the same position in the second uniformly random vector and the other third uniformly random vectors; and

calculating the first elements by applying an unmasking function to the second elements and to the m third elements.

2. The method according to claim **1**, wherein generating the third uniformly random vector ranked k comprises:

using the n first elements;

shifting the n first elements of at least k steps; and

replacing at least k elements from the first elements by k elements from the third elements.

3. The method according to claim **1**, wherein each first, second, and third elements are chosen in a finite set.

4. The method according to claim **1**, wherein the first vector is a commitment vector of a Dilithium algorithm.

5. An electronic device capable of implementing a generation method of a uniformly random first vector comprising n first elements, n being an integer equal or greater than two, each first element being comprised in a finite set T, the method comprising:

generating a second uniformly random vector comprising n second elements;

generating m third elements, m being an integer equal or greater than one; generating m third uniformly random vectors ranked from zero to m−1, wherein third uniformly random vector ranked k, k being an integer comprised between zero and m−1, comprises k third elements and n-k second elements, each third uniformly random vector having third elements different from the second elements at the same position in the second uniformly random vector and the other third uniformly random vectors; and

calculating the first elements by applying an unmasking function to the second elements and to the m third elements.

6. The device according to claim **5**, wherein generating the third uniformly random vector ranked k comprises: using the n first elements; shifting the n first elements of at least k steps; and replacing at least k elements from the first elements by k elements from the third elements.

7. The device according to claim **5**, wherein each first, second, and third elements are chosen in a finite set.

8. The device according to claim **5**, wherein the first vector is a commitment vector of a Dilithium algorithm.

9. An execution method of a cryptographic algorithm, the method comprising:

generating a second uniformly random vector comprising n second elements;

generating m third elements, m being an integer equal or greater than one;

generating m third uniformly random vectors ranked from zero to m−1, wherein a third uniformly random vector ranked k, k being an integer comprised between zero and m−1, comprises k third elements and n-k second elements, each third uniformly random vector having third elements different from the second elements at the same position in the second uniformly random vector and the other third uniformly random vectors; and

calculating the first elements by applying an unmasking function to the second elements and to the m third elements.

10. The method according to claim **9**, wherein generating the third uniformly random vector ranked k comprises:
using the n first elements;
shifting the n first elements of at least k steps; and
replacing at least k elements from the first elements by k elements from the third elements.

11. The method according to claim **9**, wherein each first, second, and third elements are chosen in a finite set.

12. The method according to claim **9**, wherein the first vector is a commitment vector of a Dilithium algorithm.

13. The method according to claim **9**, wherein the cryptographic algorithm is a signature method.

14. The method according to claim **9**, wherein the cryptographic algorithm is a Dilithium algorithm.

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